

Damien Ha

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SKILLS

Programming/Technologies Python (matplotlib, numpy, pandas, plotly, seaborn, sklearn, tensorflow, nltk), R (dplyr, ggplot2, rvest, shiny, tidyverse), C++, Tableau, SQL, Git/GitHub, html, css
Skills Data Analysis, Mathematics, Statistics, Google Suite, Microsoft Office/Excel, Teamwork, Customer Service
Languages American Sign Language (intermediate), Spanish (beginner), Vietnamese (beginner)

PROJECTS

- Fish Weight Multiple Linear Regression** <https://github.com/damienmha/Fish-Weight-MLR>
- Conduct multiple linear regression in R on fish measurement data to better understand fishery health
 - Use diagnostic plots, stepwise regression, and variable selection/transformation, to produce regression model with R^2 of 0.99
- Roller Coaster Classification** <https://github.com/damienmha/coaster-classification>
- Clean data, create data processing pipeline, do exploratory analysis, train classification models (ex: kNN, Logistic Regression, Support Vector Machines, Decision Tree) to identify coaster track type. Model performance: top accuracy and F1 scores ~ 95%
 - Train a neural network (deep learning) using Keras (TensorFlow), also to identify roller coasters by track type
- Trading Strategies (MCM 2022)** <https://github.com/damienmha/Trading-Strategies>
- Use pandas, numpy, matplotlib in Jupyter notebook to handle bitcoin/gold pricing data from *Mathematical Contest in Modeling*
 - Made mathematical models: threshold, momentum trading, moving average, and bitcoin prioritization to maximize profits
- Wordle Solver** <https://github.com/damienmha/wordlesolver>
- Program written in Python to solve partially guessed NYT Wordle puzzles based on known and potential letters and positions
 - Utilize natural language toolkit to make word dictionary, write functions to process user input, place letters, filter words, etc.

EDUCATION

University of California, Los Angeles **Sept 2020 - Sept 2025**
BS Statistics and Data Science
Relevant Coursework: Computational Statistics with R, Data Analysis and Regression, Data-Driven Mathematical Modeling, Introduction to Computer Science, Linear Algebra, Music and Data Science, Music Business Now, Principles of Accounting I, Principles of Economics, Probability and Statistics, Python for Data Science, Statistical Consulting
Activities/Societies: Association of Computing Machinery, Bruin Origami, DataRes (Data Journalist), Game Music Ensemble (clarinetist), Marching Band (clarinetist), National Student Data Corps (Board), Statistics Club, Symphonic Band (clarinetist)

CERTIFICATES AND AWARDS

LinkedIn Python Assessment, R Assessment
Kaggle Introduction to Deep Learning
ASA DataFest UCLA Finalist/Honorable Mention 2023, Best Use of External Data
California Department of Education State Seal of Biliteracy in English and American Sign Language

EXPERIENCE

DataFest at UCLA **April 2023 - April 2023**
Hackathon Finalist *Los Angeles, CA*

- Collaborate in team to analyze legal data, draw insights, allowing American Bar Association to allocate resources effectively
- Source economic data, do time series analysis, produce visualizations with Python, Tableau. Validate the use of economic identifiers, found that unemployment and legal cases have 0.658 correlation. 2nd Place in *Best Use of External Data* category
- Determine key factors in consumer finance/unemployment, advise use of regression, ARIMA models in future analyses

Northeast Big Data Innovation Hub - National Student Data Corps, UCLA **Jan 2023 - June 2024**
Board Member - Project Leader *Los Angeles, CA*

- Host social events and compile resources to facilitate a community where all students interested in data science can learn
- Assist in development of projects/workshops in data analysis, machine learning, and introductory Python for over 500 students

UCLA DataResolutions **Jan 2023 - Sept 2023**
Data Journalist *Los Angeles, CA*

- Work in teams to craft data-driven articles offering key insights into various topics, publish to Medium for public access
- Perform analysis, regression, produce visualizations with R (ggplot2), Python (pandas, matplotlib, seaborn, sklearn), Tableau

UCLA Undergraduate Math Council **Sept 2021 - Sept 2024**
Student Support Co-Director *Los Angeles, CA*

- Collect information from math students via surveys about math course content, difficulty, prerequisites, etc.
- Analyze student survey feedback to revise and provide accurate course descriptions to math students, over 400 students
- Communicate key survey information with other council members to improve student support via major maps, advice column